

- GAME DESIGN DOCUMENT -

STARFALL PROTOCOL

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Starfall Protocol - Game Design Document

GENRE: Endless Vertical Shooter ENGINE: Unity 6.6 (URP), C#

1. Pitch

Earth is under attack, and you're its last line of defense: you pilot a small ship through a scrolling starfield and fight your way through waves of enemies. Along the way you collect upgrades, grow your score, and try to beat your own high score. Easy to learn, quick to restart.

2. Design Pillars

- **Instantly readable** - enemy types, shots, and upgrades are recognizable at a glance, and the threat to Earth makes the goal immediately tangible. No tutorial needed beyond the first ten seconds.
- **Constant pressure** - difficulty ramps up continuously, never in sudden jumps. Meant to feel intense, not unfair.
- **Upgrades with real weight** - every choice is a trade-off.
- **Precise controls** - movement and shooting feel immediate. No lag, no ambiguity about whether a hit landed.

3. Core Loop

1. You start at the bottom of the screen with the base weapon.
2. Different enemies come in from above.
3. You destroy enemies and collect points.
4. On reaching a score threshold, the run briefly pauses and you pick one of 3 upgrades.
5. In the background, difficulty keeps climbing the whole time.
6. Eventually the boss appears as the run's climax.
7. Either your ship or the Earth is destroyed → the Game Over screen shows your score and your high score, and you can restart right away.

4. Mechanics

4.1 Player

- Moves left/right, constrained to the screen area.
- Fires automatically from a central firing point - you focus on positioning and dodging.
- Healthbar in the top-right corner - reduced by collisions with enemies and boss projectile hits.

4.2 Enemies (3 Types)

TYPE	BEHAVIOR	THREAT
Drifter	slow, simple movement pattern	Low
Weaver	sinusoidal, evasive movement pattern. Harder to hit	Medium
Rusher	fast, direct approach, punishes hesitation	High

All enemies run on ScriptableObjects (EnemyData) and share an IMovementPattern interface, so a new enemy type later mostly just needs a new data asset instead of an entirely new movement script.

4.3 Earth

- Visible at the bottom of the screen, with its own healthbar.
- Enemies that aren’t destroyed in time don’t just disappear but collide with Earth and deal damage.
- Damage per enemy type is tied to its speed - faster enemies are harder to intercept and deal correspondingly more damage.
- If Earth’s healthbar hits zero, it’s Game Over.

4.4 Boss

- Appears once per run as the climax.
- Two attack phases, each with its own targeted projectile pattern.
- Serves as the run’s central skill check.

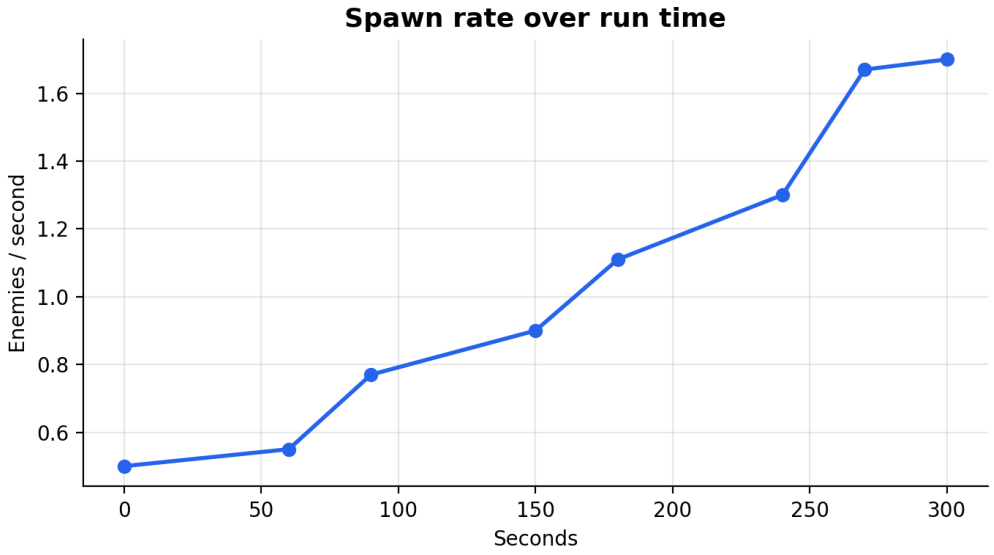
4.5 Upgrade System

- 5 upgrades total, offered as a choice of 3 at score thresholds.
- Built on a shared IUpgradeEffect interface, driven by UpgradeData ScriptableObjects, so new upgrades can be added without touching selection or application logic.
- The score threshold to the next upgrade grows after each pick, so the upgrades spread out across the run instead of clustering early and running dry later.

UPGRADE	EFFECT
Double Shot	Activates the prepared side firing points.
Fire Rate	Increases how often the ship fires.
Speed	Increases the ship's movement speed.
Shield	Each charge absorbs one hit on the ship. While at least one charge is active, a semi-transparent arc sits in front of the ship. It disappears with the last charge.
Repair	Restores 20% of maximum health. Only offered when the ship is damaged.

4.6 Difficulty Scaling

Spawn rate and enemy composition scale with survival time - no hard level boundaries, just a steady increase driven by the spawner.



Spawn rate rises continuously over the run's duration, with a noticeable jump after the boss fight.

4.7 Score

- Points increase per enemy destroyed.
- High score is saved locally (PlayerPrefs) and shown on the Game Over screen.

4.8 Playtesting & Balancing

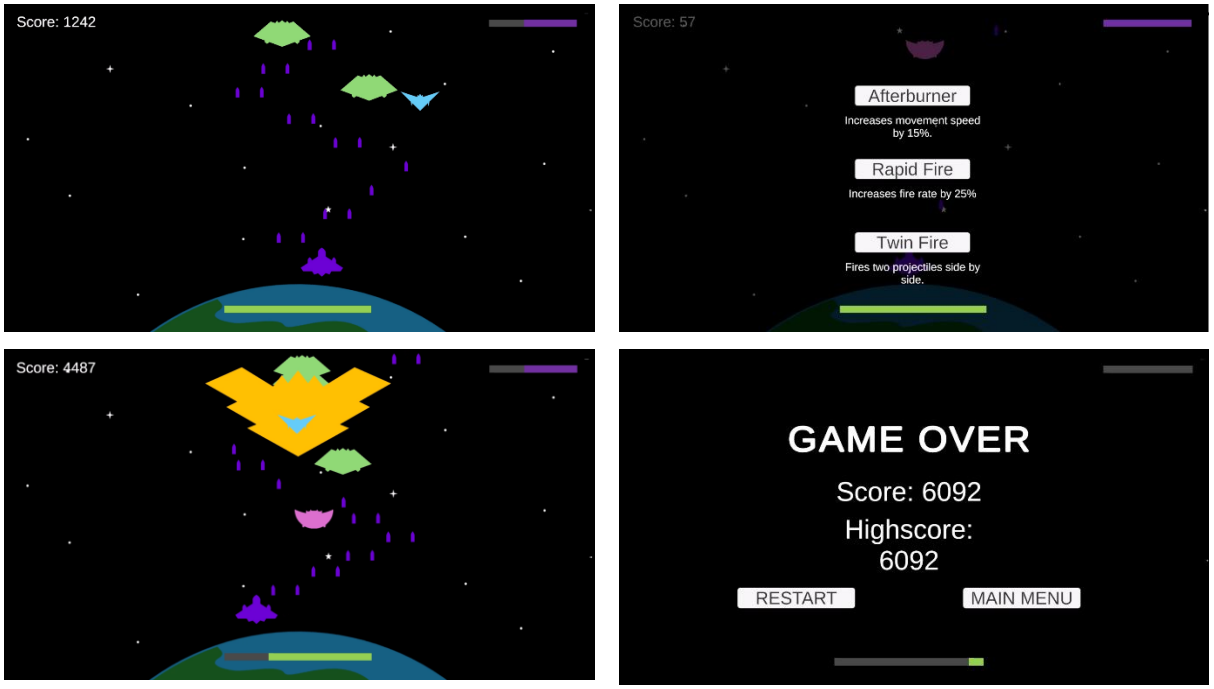
The values for enemies, weapon, and upgrades weren't set once and left alone - they were iteratively tested and adjusted, for instance when upgrades turned out too strong in early test runs and the fire-rate and speed bonuses were scaled back accordingly. The score threshold for upgrades was also changed from a fixed interval to a growing gap, so upgrades spread out across the whole run.

4.9 Audio

- **Two music loops** - a calmer one for the main menu and a driving one during runs, both looping seamlessly.
- **Crossfade** - music fades smoothly when switching between menu and game.
- **Ducking** - music is turned down while the game is paused or the upgrade selection is open.
- **Mute** - M toggles the music off and on at any time.

5. UI / UX

- **Main menu** - Start, optional Settings, Quit.
- **HUD** - player healthbar top-right, Earth healthbar on the Earth strip at the bottom, current score, high score for reference.
- **Upgrade selection screen** - appears at thresholds, shows 3 cards, pauses gameplay during selection.
- **Pause menu** - Resume, Restart, Back to menu.
- **Game Over screen** - final score, high-score comparison, restart button.
- **Mouse cursor** - hidden during gameplay and visible as soon as a menu is open (main menu, pause, upgrade selection, Game Over).



6. Artstyle

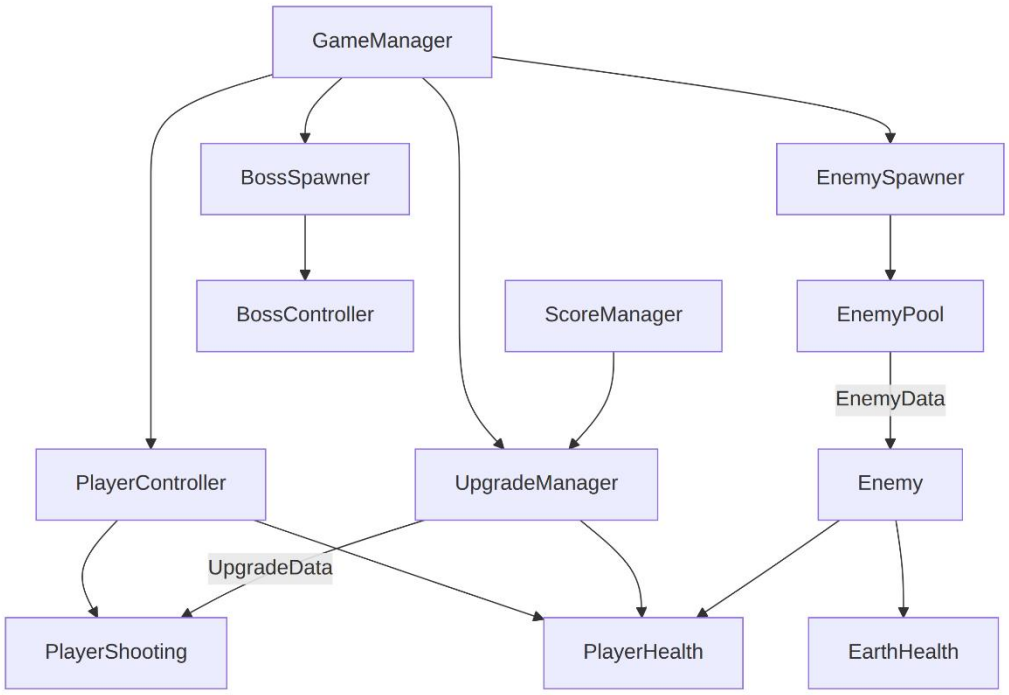
- Every visual element - ship, enemies, boss, UI, and Earth - was designed by me: modeled from geometric shapes in PowerPoint, grouped, and exported as graphics.
- High-contrast silhouettes against a scrolling starfield keep everything readable even with a lot going on.
- Colors aren’t purely decorative but make it easier to tell enemy types apart.
- The UI stays minimal so it doesn’t compete with gameplay readability.



7. Technical Structure

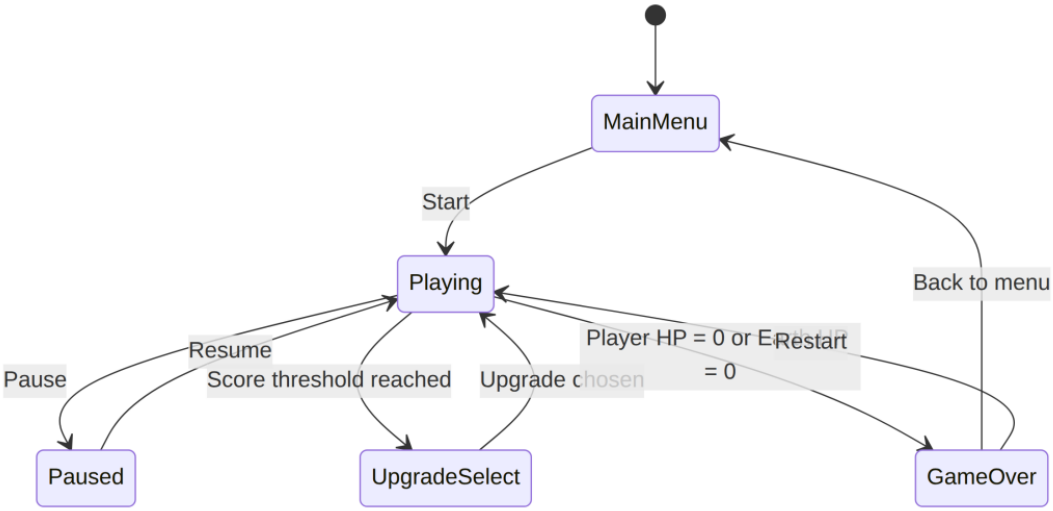
- **Object Pooling - a generic ObjectPool, plus dedicated pools for projectiles (ProjectilePool) and enemies (EnemyPool), to avoid instantiation costs during intense waves.**
- **Data-driven design** - WeaponData, EnemyData, UpgradeData, BossData as ScriptableObjects, keeping design content separate from code logic.
- **Interfaces** - IDamageable for anything that can take damage (player, Earth, enemies, boss), IMovementPattern for enemy movement, IUpgradeEffect for upgrade behavior.
- **Collision logic** - dedicated components per interaction type (enemy↔player, enemy↔Earth) instead of centralized damage dispatch, keeping responsibilities cleanly separated.
- **Systems** - PlayerController, PlayerShooting, PlayerHealth, EarthHealth, EnemySpawner, BossSpawner, ScoreManager, UpgradeManager, GameManager for pause/game-over/restart flow.

System Architecture



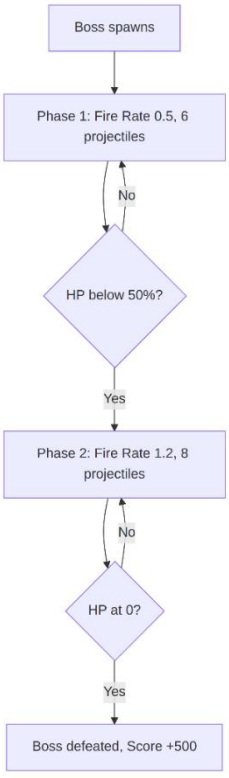
The GameManager orchestrates the subsystems; content flows in exclusively through ScriptableObjects (EnemyData, UpgradeData, BossData), never hardcoded.

Game State Flow



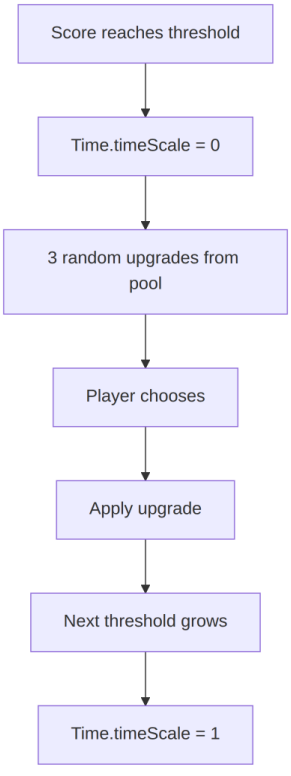
The GameManager centrally drives this state transition, including the time-stop during upgrade selection (Time.timeScale = 0).

Boss Phases



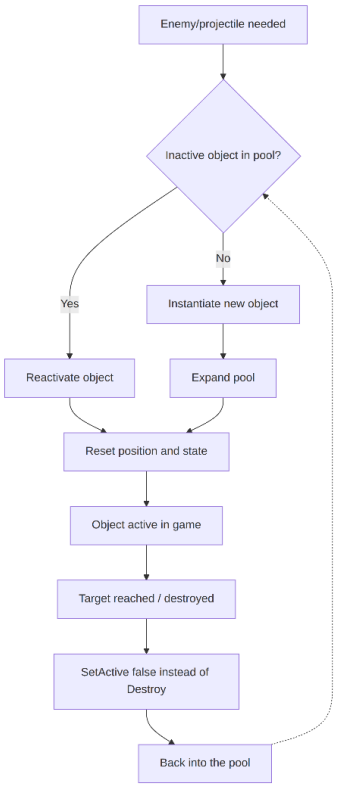
The phase change at 50% HP noticeably raises fire rate and projectile count - the boss gets denser in the second half, not just longer.

Upgrade Selection Flow



The growing threshold (see 4.5) ensures the upgrades spread out across the whole run instead of clustering early.

Object Pooling Flow



No Instantiate()/Destroy() during waves - objects are reused, which keeps the frame rate stable with a full screen of activity.

8. Scope

What's actually implemented:

3 enemy types, 1 boss, 1 base weapon, 5 upgrades, endlessly scaling difficulty, Earth and player healthbars, local high score, background music, complete menu/pause/game-over flow.

Deliberately out of scope:

Multiple weapons, multiple bosses, a level/stage structure, online leaderboards, progression between runs.

Guiding principle: finishing and polishing within the defined scope takes priority over additional scope.

9. Possible Extensions

- More enemy types and bosses.
- Meta-progression (permanent unlocks between runs).
- Online/global leaderboard.
- More weapon types selectable at the start of a run.
- Sound effects and animations.